

29 D energy needed in reaction

$$\begin{aligned}
 &= (m_{\text{H}} + m_{\text{O}} - m_{\text{N}} - m_{\alpha}) c^2 \\
 &= \left(1.007825 + 16.999130 \right. \\
 &\quad \left. - 14.003074 - 4.002604 \right) u c^2 \\
 &= 0.001277 (1.66 \times 10^{-27}) (3 \times 10^8)^2 \\
 &= 1.91 \times 10^{-13} \text{ J} = 1.19 \times 10^6 \text{ eV}
 \end{aligned}$$

30 B Isotope B could be formed from isotope Y after 2 successive alpha decays